COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_piecewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit}; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, 0.261 m	measured / CAD
	(case 1, unloaded)	30.08 N	measured
$l_{p,\phi}$, $l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	mocap arc fit
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T\Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
$\varepsilon_{\max}$	ramp-slope gate	3%	sensitivity chain (Sec. 5)
M_{floor}	slope-gate floors (x, y)	0.7, 0.4 N m	$\lvert M_{\mathrm{crit}}\rvert$ lower bound, ±20 mm box
$N_{\min}$	window-sample gate	38	30 pre + 8 post
$e_{\mathrm{lin,max}}$	linearity-RMSE gate	30 mN m	10× noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 7).

2 Pipeline overview

The complete pipeline is five stages (Fig. 1); the three algorithms nest as Algorithm 3 ⊃ Algorithm 2 ⊃ Algorithm 1, and Algorithm 1 is the only stage that touches the angular rate.

1. **Signals and window** (Sec. 4) — per run: reconstruct $M(t)$ from the measured rotor speeds and take $\omega(t)$ from odometry; open the excitation window at $\lvert M\rvert > M_{\mathrm{thr}}$ (backward from the peak), close it at the moment peak, truncated at the allocator cap.
2. **Run gates** (Sec. 5) — discard runs failing the slope-error (3%), sample-count (38) or linearity-RMSE (30 mN m) gates; moment trace only, so no circularity.
3. **Calibration, once per configuration and axis** (Sec. 7) — a per-run free fit (PNLS) runs first and its per-dataset medians locate (C_2, K); Algorithm 3 then refines that centre on a grid, and each candidate is scored by Algorithm 2, which runs Algorithm 1 on every gated run and sums the two directional CVs of M_{crit} across ramp rates.

4. **Identification, per run** (Sec. 6) — Algorithm 1 with the calibrated (C_2^*, K^*) pinned, at full sweep resolution: amplitude fixed by $C_1 = K\dot{M}$, baseline by the pre-segment median, the onset index swept exhaustively; $M_{\mathrm{crit}} = M(t_{j^*})$.

5. **Aggregation** (Sec. 8) — directional means $\bar{M}_\pm$, the pivot-free feed-forward moment $M_{\mathrm{ff}} = \frac{1}{2}(\bar{M}_+ + \bar{M}_-)$, and the CoM offset for validation against the load-cell truth (Sec. 9).

3 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P\,\dot{\omega} = M(t) + f\,l_p - W\,(l_p + \lambda_{\mathrm{off}}) + W z_{\mathrm{CoM}}\,\varphi, \tag{1}$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\mathrm{crit}}) = 0, \qquad \dot{\omega}(t_{\mathrm{crit}}) = 0. \tag{2}$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\mathrm{crit}} + \dot{M}\,\tau$ in onset-relative time $\tau = t - t_{\mathrm{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\boxed{\omega(\tau) = C_1\big(\cosh(C_2\tau) - 1\big), \qquad C_1 = K\dot{M}, \quad K = \frac{1}{W z_{\mathrm{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\mathrm{CoM}}}{J_P}}} \tag{3}$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\mathrm{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as per-configuration effective constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \to 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

4 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i}\,\Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\mathrm{end}} = \arg\max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap}: past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

5 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\mathrm{cmd}})/\dot{M}_{\mathrm{cmd}}$ and $\dot{M}$ the least-squares slope over the sub-segment $|M| \geq M_{\mathrm{floor}}$ (0.7 N m roll, 0.4 N m pitch). The floors are the per-axis lower bounds of $|M_{\mathrm{crit}}|$ over the ± 20 mm CoM-offset design box, so the gate scores the ramp exactly where an onset can occur: spin-up imperfections at near-zero moment cannot reject a run whose ramp is within spec where it matters. (On this dataset the floored gate passes all 140 runs; scored on the full window instead, 8 fast runs would be rejected by their early transients.) A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about 7× below the $\pm 20\%$ level the sensitivity analysis shows harmless.

2. **Window samples** $N_{\mathrm{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\mathrm{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\mathrm{pre}} = M_{\mathrm{crit}}^{\min}/(\dot{M}\,T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\mathrm{seg}} = 8$ post-onset fit minimum gives 38.

rate	$\lvert\varepsilon\rvert$ [%]		lin. RMSE [mN m]		N_{full}	N_{floor}
[N m/s]	med	max	med	max	min	min
0.10	0.14	1.16	2.3	3.8	484	102
0.20	0.19	1.55	1.9	3.3	257	71
0.30	0.17	0.83	1.8	2.5	191	56
0.45	0.15	2.21	1.7	2.8	130	52
0.65	0.23	0.90	1.8	2.9	95	47
0.90	0.39	2.40	1.9	2.9	77	33
1.20	0.86	1.80	2.3	4.2	53	31

Table 2: Per-rate summary of the gate quantities (20 runs per rate; slope and linearity on the floored segment). All values are inside the gates with wide margin; the mild growth of $|\varepsilon|$ toward 1.20 N m/s reflects the shorter averaging segment, not degraded execution (`analysis/gate_report.py`).

3. **Linearity RMSE** ≤ 30 mN m about the run's own best-fit line: a fault detector for aborted/stepped ramps, set at $10\times$ the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$. Figure 2 and Table 2 report the three gate quantities for all 140 runs, evaluated on the floored segment: the slope tracking error stays within $\pm 2.4\%$ (median 0.28%), the ramp-linearity RMSE within 4.2 mN m (median 2.0 — the execution noise floor itself, a factor 7 below the gate), and the shortest window carries 53 samples ($1.4\times$ the necessary minimum).

6 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\text{start}}, i_{\text{end}}]$; moment $M(t)$; calibrated (C_2, K); measured ramp slope $\dot{M}$; margin $N_{\text{seg}} = 8$
Ensure: onset index j^*; critical moment M_{crit}

$C_1 \leftarrow K\dot{M}$ ▷ onset conditions (2) fix the amplitude
for $j = i_{\text{start}} + N_{\text{seg}}, \ldots, i_{\text{end}} - N_{\text{seg}}$ **do** ▷ exhaustive sweep of the single unknown
  $C \leftarrow \text{median}\{\omega_i : i_{\text{start}} \leq i < j\}$ ▷ baseline by continuity
  $\text{cost}(j) \leftarrow \sum_{i=i_{\text{start}}}^{j-1} (\omega_i - C)^2 + \sum_{i=j}^{i_{\text{end}}} \big(\omega_i - [\,C_1(\cosh C_2(t_i - t_j) - 1) + C\,]\big)^2$
end for
$j^* \leftarrow \arg\min_j \text{cost}(j)$; $\quad M_{\text{crit}} \leftarrow M(t_{j^*})$

Two implementation details. *Baseline by median.* The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 81/140 onsets identical, but biases the remainder toward *earlier* onsets, shrinking the directional means in magnitude by up to 30 mN m on the noisiest datasets. The route is indirect. At *fixed* (C_2, K) a baseline biased toward the motion direction *delays* the detected onset: the post-onset residual $C_1[\cosh(C_2(t - t_{\text{crit}}^{\text{real}})) - \cosh(C_2(t - t_{\text{crit}}^{\text{est}}))] - C$ can only vanish with a later split, and an oppositely biased baseline advances it. The dominant effect, however, is that the leaked-rise flattening of the cost landscape moves the CV-selected calibration to a different point on the (C_2, K) ridge — e.g. $C_2 : 8.00 \to 5.38$, $K : 0.060 \to 0.150$ on the most affected dataset — whose net effect is the earlier-onset magnitude shrink; the ramp-invariance CV itself is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.0 mN m (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.5, p90 17.6, max 47.8 mrad/s), yet the estimate barely moves (66/140 onsets identical, M_{ff} shift ≤ 4.9 mN m, i.e. 0.16 mm). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a vehicle- and temperature-dependent quantity that need not stay this small — at zero parameter cost. Nor does C need the bias to be constant in time: it is re-estimated per run, so only *within-window* drift (windows last 1–10 s) matters. Measured over the guaranteed pre-onset segment ($|M| < 0.3$ N m, spans ≤ 3.0 s), the baseline wanders by 3.4 mrad/s (median; p90 24, max 68) — an upper bound that conflates true sensor drift with real elastic pre-onset motion (`analysis/bias_drift.py`). This wander is present in the very data on which the $C = 0$ bound and the rate-invariance certificates were earned, and its structure is covered by the perturbation absorption: the constant part goes to C, a τ-linear trend is absorbed by the span projection, and only the residual curvature reaches ρ. *Sweep margin.* The margin $N_{\mathrm{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\mathrm{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\mathrm{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\mathrm{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\mathrm{pre}} \geq 30$ samples inside the window (Sec. 5) and the window closes only after the response is well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a* prediction test*, not flexibility: across all* 140 *gate-passing runs the normalized post-onset residual is* 6.8% *in the median and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

The residual, measured. Table 3 carries the two normalizations that matter, both divided by the same post-onset peak excursion (`analysis/rate_residual.py`). The window RMS is the quantity of the remark above. The endpoint residual — the last three samples, averaged — is the quantity the deviation bound of the companion analysis is stated for, and it is the larger of the two because the deviation accumulates towards the end of the window.

The window is reported from the fit rather than from any design figure: C_2 is that run's own fitted α, τ_{end} the post-onset span it actually covered, and φ_{end} the integral of the fitted curve over that span — the trajectory the model claims, not the raw rate, which carries the residual itself. The dimensionless window $x = C_2\tau_{\mathrm{end}}$ then follows without solving anything.

Table 3: The window each run spanned, and its post-onset prediction residual, by ramp rate; 140 runs, 20 per rate, medians unless noted. Residual columns are normalized by that run's peak excursion. C_2 is a per-configuration constant, not a per-run one, so it does not vary down the table; across the ten configurations it runs 3.50 to 8.00 s^{-1}.

$\dot{M}$	τ_{end}	$x = C_2\tau_{\mathrm{end}}$	φ_{end}	peak	endpoint [%]		RMS [%]	floor [%]
[N m/s]	[s]	med (p10–p90)	[deg]	[rad/s]	median	p90	median	median
0.10	0.720	3.47 (2.9–5.0)	4.43	0.447	8.77	19.24	6.19	3.34
0.20	0.610	2.90 (2.4–4.2)	4.33	0.460	9.67	15.35	6.58	3.16
0.30	0.545	2.64 (2.0–4.0)	4.67	0.515	8.22	16.68	7.12	2.92
0.45	0.525	2.33 (1.9–3.7)	5.24	0.607	8.67	16.59	6.27	2.40
0.65	0.470	2.15 (1.6–3.5)	5.07	0.630	9.14	15.93	7.00	2.20
0.90	0.430	1.90 (1.5–3.2)	5.37	0.658	12.59	19.97	7.55	2.12
1.20	0.405	1.79 (1.4–3.0)	5.22	0.702	10.67	16.30	7.19	1.73
all	0.520	2.67 (1.7–4.0)	4.90		9.19	17.64	6.75	2.44

Two things about the window itself. The runs stop well inside the 10° design limit, at 4.9° in the median and 0.9–7.8° over individual runs, because the excitation window closes on the moment rather than on the tilt: x runs 3.47 down to 1.79 across the sweep where the tilt limit alone would give 5.18 down to 2.99. And the fitted C_2 does *not* sit at the geometric ceiling $\sqrt{g z_{CoM}/(z_{CoM}^2 + (l_p + p_{\mathrm{off}})^2)}$ of 5.05 on roll and 5.25 on pitch. Its median across configurations lands at 5.06, but that is an accident of the median: the ten values are 3.50, 3.75, 3.88, 3.88, 4.50, 5.63, 6.13, 6.63, 6.88 and 8.00, and the largest exceeds the ceiling by 59% — which a ceiling forbids. Either p_{off} does not belong in the arm of that bound, or the contact is not the rigid pivot the bound assumes. Nothing downstream rests on it: C_2 enters the identified threshold only through the onset index, and Sec. 9 shows the seven detectors agreeing to 3–45 mN m regardless.

Before reading the residual it is worth saying what the fit leaves free, because the usual objection to a fit residual — that least squares drives it orthogonal to the model and so flatters it — does not apply here. Nothing in this fit is a continuous parameter: C_1 is pinned to $K\dot{M}$, C_2 is the configuration's constant, the baseline is a median rather than a least-squares estimate, and only the integer onset index is searched. The one stationarity that does exist is in that index, and it is the orthogonality $\langle \text{residual}, \chi \rangle = 0$ with $\chi = \partial\hat{\omega}/\partial t_c \propto \sinh C_2\tau$. So the fit absorbs whatever part of the deviation looks like a shifted onset, which is the part concentrated *early*: a disturbance concentrated at the end of the window is not absorbed and does reach the residual.

Three readings. *It is flat in rate*, 8.77% over the slow half ($\dot{M} \leq 0.30$) against 10.06% over the fast half (≥ 0.65), which is the certificate the remark claims and the reason no ramp rate has to be preferred. *It is not gyro noise*: in absolute terms the residual is 0.0452 rad/s against a pre-onset floor of 0.0148, a factor of 3.0, and it stays near that level while the peak excursion varies sixfold across the runs (log–log slope 0.40, $R^2 = 0.02$).

And it behaves like a growth-rate mismatch. Because C_2 is pinned, a discrepancy in it survives into the residual as $\partial\omega/\partial C_2 = C_1\tau \sinh C_2\tau$, whose endpoint value relative to the peak is exactly $\Psi(x) = x \coth(x/2)$. That signature is testable, and it holds: across configurations at a fixed rate the rank correlation between $\Psi(x)$ and the endpoint residual is $+0.379$ ($p = 0.010$), while across rates within one configuration it is -0.051 ($p = 0.73$). The residual is therefore a property of the configuration and not of the excitation, which is also why it is flat in rate — C_2 is a configuration constant. Read that way it implies a residual $|\delta C_2/C_2|$ of 3.0% in the median and 5.5% at p90. That is not the same as saying the fitted C_2 is wrong by its departure from the geometric ceiling above: a mismatch of that size would leave a residual near 75%, not 9%.

And it is not the modelled forcing either, which is worth establishing rather than assuming. The deviation bound predicts a residual falling five-fold from the slowest ramp to the fastest, since the moment swept before the tilt limit is reached collapses as $\dot{M}^{2/3}$; Table 3 shows no such fall. The test that settles it is whether the runs the bound calls worse are in fact worse. Within each rate, so that the rate trend cannot carry the comparison, the rank correlation between the predicted bound and the measured residual is -0.198 ($p = 0.005$ across the seven rates) — the wrong sign — while against the pre-onset floor it is $+0.200$. The modelled forcing is therefore below the floor everywhere, and the bound should be read as a guarantee on one channel rather than as a description of what the runs do. Note also that the runs stop at $4.9°$ of tilt rather than at the $10°$ design limit, so a bound stated at that limit overstates the gravity channel by roughly the square of the ratio.

None of this reaches the deliverable, for the reason given in Sec. 10: the onset is located by a normalized correlation, so a disturbance uncorrelated with its weight averages out. A residual of 9% on the rate leaves the identified threshold displaced by 0.04 mN m in the median and 0.15 at worst.

7 Calibration of the effective constants

Algorithm 1 presupposes (C_2, K). Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the attitude gradient of the ground effect). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion in practice. On the fitted window the amplitude and the onset are nearly interchangeable under noise: a smaller amplitude with an earlier onset and a larger amplitude with a later onset produce post-onset curves that differ most near the onset itself, where the response is quadratic, $\frac{1}{2}C_1C_2^2\tau^2$, and of the order of the gyro noise. In the noiseless limit the post-onset shape determines all three parameters uniquely — the trade-off is an ill-conditioning statement, not a non-identifiability one — but under gyro noise the summed residual is nearly flat along this amplitude–onset direction: a *shallow ridge* in the (C_2, K) plane, observed directly on the data (the objective is piecewise-constant in the integer onset index, gradient-based and simplex optimizers stall on it, and a population-based global search costs about 4× more for no measurable gain). Per-run estimates — and with them the detected critical moments — scatter along the ridge. The calibration should therefore be read as *variance suppression*: choosing (C_2, K) once per dataset to minimize the dispersion of M_{crit} across ramp rates removes precisely that scatter, which is where the practical strength of the method lies.

The scatter direction is faint in the residual but loud in the deliverable: moving along the trade-off shifts each detected onset in *time*, and since $M_{\text{crit}} = M(t_{\text{on}})$ with M ramping at $\dot{M}$, an onset-time bias δt common across runs biases the critical moment by $\dot{M}\,\delta t$ — growing linearly with the ramp rate. A wrong pair thus *fans out* the identified $M_{\rm crit}$ across the twelvefold rate range, while the correct pair leaves it rate-invariant: exactly the direction a rate-consistency criterion sees best. We therefore calibrate by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \dots, 1.20\}$ N m/s,

$$\mathrm{score}(C_2, K) \;=\; \sum_{\mathrm{dir}\in\{+,-\}} \mathrm{CV}\Big(\big\{M_{\rm crit}^{(r)}(C_2,K)\big\}_r\Big), \qquad \mathrm{CV} = \mathrm{std}/|\mathrm{mean}|, \tag{4}$$

where $M_{\rm crit}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r.

Search ranges. The stage-2 grid is centred on the stage-1 medians and spans $\pm 25\%$ in C_2 and $[-40\%, +40\%]$ in K, which is wide enough to absorb the scatter of the per-run fits (interquartile ranges of roughly $\pm 10\%$ in C_2) without reaching the unphysical extremes a fixed box admits. For reference, the earlier fixed box was $K \in [0.10, 0.50]$ $(\mathrm{N\,m})^{-1}$, i.e. $z_{\rm CoM} \in [0.06, 0.32]$ m at the vehicle weight, with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ ($z_{\rm CoM}$ up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(KC_2^2)$ spans roughly $[0.03, 1.1]$ kg m^2, bracketing the parallel-axis estimate $J + m z_{\rm CoM}^2 \approx 0.2$–$0.35$ kg m^2 by a factor of several on both sides.

Algorithm 2 Ramp-invariance consistency score

Require: set $\mathcal{G}$ of gated runs (both tip directions, spanning the rate set $\{0.10, \dots, 1.20\}$ N m/s); candidate pair (C_2, K)
Ensure: consistency score s ▷ lower $=$ $M_{\rm crit}$ more rate-invariant
for dir $\in \{+,-\}$ **do**
  $\mathcal{G}_{\rm dir} \leftarrow \{\rho \in \mathcal{G} : \mathrm{dir}(\rho) = \mathrm{dir}\}$
  for each run $\rho \in \mathcal{G}_{\rm dir}$ **do**
    $M_{\rm crit}^{(\rho)} \leftarrow$ Algorithm 1 with (C_2, K) pinned ▷ stride-3 sweep
  end for
  $\mathrm{CV}_{\rm dir} \leftarrow \mathrm{std}\big(\{M_{\rm crit}^{(\rho)}\}_{\rho\in\mathcal{G}_{\rm dir}}\big) \,/\, \big|\mathrm{mean}\big(\{M_{\rm crit}^{(\rho)}\}_{\rho\in\mathcal{G}_{\rm dir}}\big)\big|$
end for
return $s \leftarrow \mathrm{CV}_+ + \mathrm{CV}_-$ ▷ Eq. (4): a static threshold must not depend on the approach rate

Note that the ramp rate never appears explicitly in the score: each direction's gated runs span the full rate set, so the within-direction dispersion of the per-run critical moments *is* the rate dependence (plus repeatability scatter, which the score harmlessly co-penalizes). Should a rate be repeated, each run enters individually.

Algorithm 3 Effective-constant calibration (PNLS-located, then refined)

Require: gated runs of one dataset
Ensure: calibrated pair (C_2^*, K^*)
locate (PNLS, runs first): fit (C_1, C_2, C) per run by free nonlinear least squares; record C_2 and $K = C_1/\dot{M}$
$(C_2^0, K^0) \leftarrow$ per-dataset medians ▷ owes nothing to the ramp-invariance criterion
refine: evaluate Algorithm 2 on the 9×9 grid $C_2 \in C_2^0[0.75, 1.25]$, $K \in K^0[0.60, 1.40]$
$(C_2^*, K^*) \leftarrow \arg\min$ over that grid

Two-stage procedure. The two stages consume independent information. The PNLS pass reads the *shape* of each run's post-onset curve and is blind to how the threshold varies with ramp rate; the refinement reads exactly that variation and is nearly blind to the combination $KC_2^2 = 1/J_P$ that the shape fixes (Remark 2). Running PNLS first and refining about its medians keeps the search off the arbitrary edges of a fixed box: the refined constants land at $C_2 \in [3.52, 5.67]$ rad/s and $K \in [0.150, 0.338]$ $(\mathrm{N\,m})^{-1}$, whereas minimising the same score over the wide box $[3, 8] \times [0.05, 0.70]$ puts case $1/M_x$ on the upper bound at $C_2 = 8.000$ and spreads K over 0.060–0.520, a factor of 8.7. Accuracy and consistency are unaffected — RMS 1.65 mm against 1.64, median CV 2.5% against 2.5% (Tables 8 and 11) — so the gain is interpretability, not performance. Three of the twenty refined coordinates still sit on their local grid edge (case $3/M_x$ in C_2, cases 4 and $5/M_y$ in K), where the score genuinely prefers to move away from the shape-based centre.

Grid resolution. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. Stage 2 evaluates (4) on a 9×9 uniform grid about the stage-1 centre, i.e. resolutions of about 0.13 rad/s in C_2 and $0.05\,K^0$ in K (81 evaluations per dataset). One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about $4\times$ more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Why rate invariance is the right criterion, and not merely a convenient one. The usual justification — that joint residual minimization is degenerate along a ridge, so an independent physical fact is needed to break it — says why *some* extra criterion is required, not why *this* one. The sharper statement is that rate invariance is the matched estimator for the error the constants can commit.

Suppose C_2 is mis-specified by δC_2. The predicted curve is then wrong by $\partial\omega/\partial C_2 = C_1\tau \sinh C_2\tau$, and the onset sweep converts that into a shift by the same projection that governs every other perturbation, $\delta t_c = \langle \cdot, \chi\rangle/\|\chi\|^2$ with $\chi \propto \sinh C_2\tau$. Both carry $\sinh C_2\tau$, so the projection reduces to a weighted mean of τ and the amplitude cancels:

$$\delta t_c = -\frac{\delta C_2}{C_2}\,\langle\tau\rangle, \qquad \langle\tau\rangle = \frac{\int_0^{\tau_{\text{end}}} \tau \sinh^2 C_2\tau\,\mathrm{d}\tau}{\int_0^{\tau_{\text{end}}} \sinh^2 C_2\tau\,\mathrm{d}\tau} \approx \tau_{\text{end}} - \frac{1}{2C_2}, \tag{5}$$

the approximation holding to better than 7% over the windows these runs span, and to 0.4% at the slowest ramp. Carrying it through $\Delta M_{\text{crit}} = \dot{M}\,\delta t_c$, a growth-rate error costs $\dot{M}\,\tau_{\text{end}} \approx \Delta M_{\text{win}}$ times $\delta C_2/C_2$ — that is, it displaces the threshold *in proportion to the moment the window spans*, and therefore not at all in the same way at two different ramp rates. At $\delta C_2/C_2 = 3\%$ it is worth 1.9 mN m at the slowest ramp and 11.8 at the fastest.

That is the whole argument. A mis-specified growth rate is invisible to a per-run residual, which can trade it against the onset, but it cannot hide from a comparison *across* ramp rates, because the trade it needs is rate-dependent. Minimizing the dispersion of M_{crit} over the sweep is thus, to first order, minimizing $|\delta C_2|$ against the one channel δC_2 contaminates. It also fixes what the certificate does and does not show: the low dispersion of Table 4 is the objective's own value and cannot be quoted back as evidence for rate invariance (`analysis/rate_invariance.py` separates the two).

What survives calibration is the part of $\partial\omega/\partial C_2$ that no onset shift can represent, and that is what the prediction residual of Sec. 6 measures: 3.0% in the median. Feeding that back through (5) predicts a residual rate trend in M_{crit} of 9.9 mN m across the sweep, against a measured 10.3 ± 7.7 — consistent, and in both readings far below the 57 mN m the offset accuracy of Sec. 10 corresponds to.

What the criterion buys. Table 4 reports the quantity (4) is built from: the coefficient of variation of $|M_{\text{crit}}|$ within a tip direction, over a rate sweep spanning a factor of 12.4 ($\dot{M} = 0.10$ to 1.24 N m/s, 14 runs per dataset, 140 runs in total). The calibrated constants hold the threshold to 2.4% in the mean and 4.2% at worst. The criterion is not flat in K: halving it about the CAD point nearly doubles the dispersion, to 7.9% in the mean and 14.6% at worst. Doubling it instead *improves* the CAD figures, to 3.1% and 4.8% — the calibration's own preference showing through, since every fitted K on the pitch axis exceeds the CAD value. That asymmetry is the disagreement of Table 5 seen from the objective's side.

Remark 2 (What the valley does and does not determine) *The objective* (4) *retains a shallow valley, and the two constants are* not *equally determined along it. Read literally,* $1/K$ *would be* $W z_{\text{CoM}}$*; across the ten datasets it implies* z_{CoM} *between* 0.061 *and* 0.554 *m, and the roll and pitch datasets of* the same airframe *disagree by up to a factor* 8 *(case 1:* 0.554 *versus* 0.068 *m). Since only horizontal masses are moved between configurations,* z_{CoM} *is common across them, so* $1/K$ *is better read as an effective gain than as a CoM height. The combination* $J_P = 1/(KC_2^2)$ *behaves quite differently:* 0.249 ± 0.016 *kg m*^{2} *over the five roll datasets* ($\text{CV} = 6.5\%$) *and* 0.140 ± 0.034 *over the five pitch datasets, against* $\text{CV} = 65\%$ *and* 49% *for* $1/K$*.*

Table 5 makes the comparison explicit against a fully independent reference. The CAD model gives $z_{\text{CoM}} = 0.261$ *m — measured from the landing-gear contact plane, so it is already referred to the pivot and needs no datum correction — together with the CoM inertias of Table 5, so the parallel-axis theorem*

Table 4: Coefficient of variation of $|M_{\text{crit}}|$ within a tip direction [%], averaged over the two directions. *Free*: (C_2, K) calibrated per dataset (twenty free numbers). *CAD*: constants derived from $z_{\text{CoM}} = 0.261$ m and the Table 5 inertias by the parallel-axis theorem, with nothing fitted. Last two columns: K mis-specified about the CAD point.

Case	Axis	Free	CAD	$K/2$	$2K$
1	M_x	4.2	5.2	8.7	4.7
1	M_y	2.5	6.9	14.6	2.9
2	M_x	0.9	2.7	5.8	2.0
2	M_y	1.4	5.6	10.3	2.2
3	M_x	1.8	1.9	4.6	3.0
3	M_y	1.8	3.2	7.5	2.2
4	M_x	2.4	2.6	4.7	3.2
4	M_y	2.9	4.6	8.1	3.2
5	M_x	2.6	3.5	6.4	3.0
5	M_y	3.8	5.6	9.7	3.9
mean		2.4	4.2	7.9	3.1
worst		4.2	6.9	14.6	4.8

fixes the pivot inertia without any fit, $J_P = J_{\text{CAD}} + m(z_{\text{CoM}}^2 + l_p^2) = 0.334$ *and* 0.309 *kg m*^{2}*, and with it both constants:* $C_2 = 4.97$ *and* 5.17 *rad/s,* $K = 0.121$ *(N m)*$^{-1}$*. The products of inertia are four orders of magnitude below the diagonal, so treating the two axes independently is justified.*

The pivot arm entering that expression is measured, not assumed. The mocap marker traces an arc about the ground contact during the tip-over, and `estimate_pivot_from_mocap` *fits it with the circle centre held at ground level, leaving the horizontal offset and the radius free. Over the* 138 *runs with a valid arc the fit returns* $l_p = 0.1404 \pm 0.0036$ *m on roll and* 0.1126 ± 0.0066 *m on pitch, against the* 0.140 *and* 0.110 *m used throughout (*-0.3% *and* -2.3%*), with a circle residual of* 0.1*–*0.2 *mm. The same fit puts the marker at* 0.3172 ± 0.0012 *m above the pivot, matching the* $\sqrt{R^2 - l_p^2} = 0.3167$ *m implied by the radius and the rotor-hub height* $h = 0.315$ *m used in the ground-effect model. The geometry is therefore consistent to about a millimetre: the mocap heights are referred to the ground contact, the CAD height is referred to the landing gear, and the CAD CoM lands* 56 *mm below the marker plane, as it should for an airframe carrying its battery low. Both geometric inputs to the parallel-axis floor are thus independently confirmed, and the identified* J_P *is the only quantity left that can account for the gap.*

The disagreement is not spread evenly over the (C_2, K) *plane, and that is the substance of the remark. The constant the deliverable is sensitive to,* C_2*, agrees: the per-run nonlinear least squares gives a median of* 4.765 *rad/s,* 4.2% *and* 7.8% *below the CAD values, and the calibrated means are within* 14%*. The scale of the pair does not: read literally the calibration puts* J_P *at* 0.249 *and* 0.140 *kg m*^{2}*, which are* 25% *and* 55% *below CAD and — more tellingly —* 12% *and* 46% *below the parallel-axis floor* $m(z_{\text{CoM}}^2 + l_p^2)$ *obtained by setting* J_{CAD} *to zero.* The identified inertia is not physically attainable. *Since* $C_2^2 = W z_{\text{CoM}}/J_P$ *is confirmed while* J_P *is not, what the calibration fails to resolve is the scale of the pair, not their ratio — exactly the* K *direction that the log sensitivities below identify as the weak one.*

Table 5: Calibrated constants against the CAD reference ($z_{\text{CoM}} = 0.261$ m, Table 5 inertias, parallel-axis theorem; nothing fitted). CV and spread are over the five datasets of the axis. The last row of each block also gives the parallel-axis floor, the value J_P would take if the CoM inertia were zero.

Axis	Quantity	mean	CV	spread	CAD	difference
Roll	C_2 [rad/s]	5.65	34.8%	2.29×	4.97	+13.6%
	$1/K = W z_{\text{CoM}}$ [N m]	8.71	64.7%	5.00×	8.25	+5.6%
	$J_P = 1/(K C_2^2)$ [kg m^2]	0.249	6.5%	1.16×	0.334	−25.3%
	(floor 0.283)					−11.8%
Pitch	C_2 [rad/s]	4.90	24.5%	1.71×	5.17	−5.2%
	$1/K = W z_{\text{CoM}}$ [N m]	3.46	49.2%	2.89×	8.25	−58.0%
	$J_P = 1/(K C_2^2)$ [kg m^2]	0.140	24.1%	1.95×	0.309	−54.7%
	(floor 0.258)					−45.8%

Pitch is the weaker axis throughout, and its error is systematic rather than scattered: all five pitch datasets put $1/K$ *below the CAD value (*1.92 *to* 5.56 *against* 8.25 *N m), whereas the roll datasets bracket it. The excitation window is shortest on that axis —* τ_{end} *has a median of* 0.450 *s against* 0.555 *s for roll, and falls to* 0.23 *s at the highest rates — so* $x = C_2 \tau_{\text{end}}$ *is smallest there and the curvature information* *that separates J_P from $W z_{\text{CoM}}$ is correspondingly weakest.*

This is structural, not accidental. At the window edge, with $x = C_2\tau_{\text{end}}$, the log sensitivities are

$$\frac{\partial \ln \omega}{\partial \ln C_2} = x \coth \frac{x}{2} \;\geq\; 2, \qquad \frac{\partial \ln \omega}{\partial \ln K} = 1,$$

so the two cancel along $\delta \ln K = -x \coth(x/2)\, \delta \ln C_2$, which for small x is exactly $\delta \ln K = -2\, \delta \ln C_2$, i.e. $KC_2^2 = 1/J_P$ constant. The valley therefore runs along the direction in which C_2 carries twice the leverage of K; equivalently, near the onset $\omega \simeq (\dot{M}/J_P)\tau^2/2$ depends on the pair only through J_P. The measured spread confirms the factor: over the search box $1/K$ moves by $2.36\times$ *while C_2 moves by* $1.54\times$, *and* $\ln 2.36/\ln 1.54 = 1.99$.

Robustness certificate: three constants for ten datasets. Because the individual constants are weakly determined, it is reasonable to ask whether (C_2, K) are doing the work of a two-parameter curve family rather than of the geometry they name. A direct test is to remove the freedom. Physics requires z_{CoM} to be shared by the whole vehicle and J_P to be shared by all datasets on a given axis, so we reparametrize by

$$\big(W z_{\text{CoM}},\ J_{P,\text{roll}},\ J_{P,\text{pitch}}\big), \qquad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_{P,\text{axis}}}}, \qquad K = \frac{1}{W z_{\text{CoM}}}, \tag{6}$$

minimize the *same* score (4) summed over all ten datasets, and rerun the identification. Twenty free numbers become three; the roll and pitch inertias decouple, so the inner search is two independent scalar minimizations per candidate $W z_{\text{CoM}}$. The optimum is $W z_{\text{CoM}} = 5.50$ N m ($z_{\text{CoM}} = 0.174$ m), $J_{P,\text{roll}} = 0.240$ and $J_{P,\text{pitch}} = 0.153$ kg m^2, giving $C_2 = 4.79$ and 6.00 rad/s and $K = 0.182$ (N m)$^{-1}$ — close to the representative values quoted throughout. Table 6 compares the delivered CoM offsets.

Table 6: Identified CoM offset [mm] against the load-cell truth, with the constants calibrated per dataset (twenty free numbers) and with the physically shared parametrization (6) (three).

Case	Axis	Truth	Free	error	Constrained	error
1	M_x	−2.90	−1.73	+1.17	−1.64	+1.26
1	M_y	−11.45	−11.13	+0.32	−11.25	+0.20
2	M_x	−14.29	−15.17	−0.88	−15.16	−0.87
2	M_y	−9.90	−10.49	−0.59	−10.53	−0.63
3	M_x	−5.26	−8.62	−3.36	−8.66	−3.40
3	M_y	3.14	0.73	−2.41	0.63	−2.51
4	M_x	6.67	5.26	−1.41	5.23	−1.44
4	M_y	2.40	0.16	−2.24	0.11	−2.29
5	M_x	10.91	10.70	−0.21	10.79	−0.12
5	M_y	−10.89	−10.73	+0.16	−10.62	+0.27
RMS error				1.64		1.67
max |error|				3.36		3.40
mean error				−0.95		−0.95

Collapsing the twenty free numbers to three changes the RMS error by 0.03 mm and no individual case by more than 0.11 mm. The identical mean error, −0.95 mm in both columns, further shows that the residual bias is a property of the vehicle–contact geometry, common to every estimator, and not of the calibration. In the constrained form C_2 is also derived rather than searched, so it cannot settle on a search bound.

One caution completes the picture: this does *not* measure z_{CoM}. The score varies by only 3.6% across the whole box $z_{\text{CoM}} \in [0.15, 0.35]$ m, so every physically plausible height performs comparably. The identification therefore does not depend on knowing z_{CoM}, which is also why the individual constants are best reported as effective values rather than as geometry.

The same test with no fitted number at all. The three-constant form above still minimizes (4). The strictest variant fits nothing: CAD supplies z_{CoM} and the CoM inertias, the parallel-axis theorem supplies J_P, and both constants follow. Table 7 reruns the identification on that point.

The RMS error is 1.64 mm either way. Twenty fitted numbers and none deliver the same offset, and the mean error stays near −0.9 mm in both — a residual belonging to the vehicle–contact geometry, not

Table 7: Identified CoM offset [mm] with the constants taken entirely from CAD — $z_{\mathrm{CoM}} = 0.261$ m, Table 5 inertias, no quantity fitted to anything. *CV*: rate consistency of $|M_{\mathrm{crit}}|$ as in Table 4.

Case	Axis	J_P	C_2	K	CV [%]	Truth	Identified (error)
1	M_x	0.320	4.95	0.127	5.2	−2.90	−1.61 (+1.29)
1	M_y	0.297	5.15	0.127	6.9	−11.45	−11.16 (+0.29)
2	M_x	0.334	4.97	0.121	2.7	−14.29	−15.16 (−0.87)
2	M_y	0.309	5.17	0.121	5.6	−9.90	−10.37 (−0.47)
3	M_x	0.334	4.97	0.121	1.9	−5.26	−8.68 (−3.42)
3	M_y	0.309	5.17	0.121	3.2	3.14	0.74 (−2.40)
4	M_x	0.334	4.97	0.121	2.6	6.67	5.24 (−1.43)
4	M_y	0.309	5.17	0.121	4.6	2.40	0.34 (−2.06)
5	M_x	0.334	4.97	0.121	3.5	10.91	10.78 (−0.13)
5	M_y	0.309	5.17	0.121	5.6	−10.89	−10.34 (+0.55)
CAD only (nothing fitted)						RMS 1.64, max 3.42, mean −0.87	
Free calibration (twenty fitted numbers)						RMS 1.64, max 3.36, mean −0.95	

to the calibration. What the fitting does buy is rate consistency: the CV rises from 2.4% to 4.2% in the mean and from 4.2% to 6.9% at worst when the constants are taken from CAD.

This is the sharpest available statement of what the constants are for. The calibration cannot be measuring the rig geometry — its J_P is below the parallel-axis floor — yet the geometry it fails to recover can be substituted wholesale without moving the deliverable. Both facts follow from the same structure: the identified threshold responds to C_2, which the two routes agree on, and only weakly to K, which is where they part.

8 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\mathrm{ff}} = \tfrac{1}{2}\,(\bar{M}_+ + \bar{M}_-), \tag{7}$$

in which the pivot-arm terms $\pm(W-f)\,l_p$ — and any thrust-proportional contamination acting through the arm l_p, ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight. A roll excitation (M_x) senses the y-offset via $M_{\mathrm{ff},x} = +W y_{\mathrm{off}}$; a pitch excitation ($M_y$) senses the x-offset via $M_{\mathrm{ff},y} = -W x_{\mathrm{off}}$.

9 Comparison with per-run estimators: NLS, CPD, CUSUM

Six alternatives were run on *identical* inputs — same signals, excitation windows, allocator caps and onset-free gates (the gated run set is method-independent because the gates use the moment trace only), 140 runs in total (`analysis/nls_comparison.py`):

- **COSH**$_{\mathrm{wide}}$, **COSH**$_{\mathrm{CAD}}$ — the same detector with the constants chosen differently: the score minimised over the wide box $[3,8]\times[0.05,0.70]$ instead of around the PNLS centre, and the constants derived from CAD geometry with nothing fitted (Sec. 7);
- **NLS** — the per-run stage-1 fit used on its own: nonlinear least squares (SCIPY `least_squares`, trust-region reflective) for free (C_1, C_2, C), with C_2 bounded to $[3,8]$ rad/s and the onset swept locally around a quadratic-model seed;
- **CPD**$_{\mathrm{N}}$ / **CPD**$_{\mathrm{R}}$ — optimal single change-point detection (`ruptures`; with one break the search over all admissible splits is exact) under the Gaussian mean+variance cost and the nonparametric RBF-kernel cost, respectively;
- **CUSUM** — the tabular sequential detector with allowance $k = 2\sigma$ against the pre-onset vibration.

Figure 3 is the readable summary: per case–axis pair it shows six estimators' CoM offset with its 95% confidence interval (Welch t on the pivot-free average, run-to-run scatter) against the load-cell truth. The

case	axis	λ_{truth} [mm]	estimated CoM offset $\hat{\lambda} \pm$ CI$_{95}$ [mm] COSH	COSH$_{\mathrm{wide}}$	COSH$_{\mathrm{CAD}}$	NLS	CPD$_{\mathrm{N}}$	CPD$_{\mathrm{R}}$	CUSUM
1	x	−2.90	−1.65 ±0.94	−1.73 ±0.93	−1.61 ±1.02	−2.05 ±1.30	−1.46 ±1.03	−1.20 ±1.51	−1.15 ±1.24
1	y	−11.45	−11.18 ±0.42	−11.13 ±0.43	−11.16 ±0.78	−11.35 ±1.37	−11.21 ±0.61	−10.79 ±0.80	−9.59 ±3.88
2	x	−14.29	−15.15 ±0.36	−15.17 ±0.36	−15.16 ±0.64	−15.03 ±0.76	−15.09 ±0.70	−14.10 ±4.15	−15.00 ±0.52
2	y	−9.90	−10.57 ±0.41	−10.49 ±0.42	−10.37 ±0.80	−9.60 ±1.13	−9.91 ±0.52	−9.61 ±0.46	−8.54 ±4.02
3	x	−5.26	−8.60 ±0.45	−8.62 ±0.46	−8.68 ±0.48	−8.54 ±0.74	−8.11 ±1.20	−4.63 ±4.72	−8.45 ±0.92
3	y	+3.14	+0.72 ±0.40	+0.73 ±0.39	+0.74 ±0.60	+1.06 ±0.97	+0.71 ±0.82	+0.94 ±1.00	+1.12 ±3.39
4	x	+6.67	+5.24 ±0.65	+5.26 ±0.65	+5.24 ±0.66	+4.94 ±0.88	+5.57 ±0.72	+3.58 ±2.19	+5.01 ±0.91
4	y	+2.40	+0.14 ±0.61	+0.16 ±0.61	+0.34 ±0.79	−0.64 ±1.49	−0.66 ±1.39	−0.60 ±1.36	−0.22 ±1.36
5	x	+10.91	+10.78 ±0.73	+10.70 ±0.69	+10.78 ±0.80	+10.05 ±1.10	+10.29 ±0.65	+11.16 ±3.75	+10.25 ±0.74
5	y	−10.89	−10.57 ±0.68	−10.73 ±0.70	−10.34 ±0.93	−11.11 ±1.68	−10.84 ±0.98	−10.95 ±0.95	−10.84 ±1.00
component RMS [mm]			1.65	1.64	1.64	1.72	1.67	1.65	1.82
max |component| [mm]			3.34	3.36	3.42	3.28	3.06	3.09	3.19
mean component [mm]			−0.93	−0.95	−0.87	−1.07	−0.91	−0.46	−0.58
radial RMS [mm]			2.33	2.31	2.32	2.43	2.36	2.34	2.57
radial max [mm]			4.12	4.14	4.18	3.89	3.74	4.31	3.77

Table 8: Identified CoM offset against the load-cell truth, on identical inputs (140 gated runs). **COSH** is the reported estimator, its constants calibrated in a neighbourhood of the per-run PNLS medians; COSH$_{\mathrm{wide}}$ minimises the same score over the box $[3, 8] \times [0.05, 0.70]$; COSH$_{\mathrm{CAD}}$ derives the constants from CAD geometry with nothing fitted. A roll excitation (M_x) senses y_{off} and a pitch excitation (M_y) senses x_{off}, so each case contributes both components; the *radial* rows combine them per case as $\sqrt{e_x^2 + e_y^2}$.

case	axis	truth [mN m]	feed-forward compensation moment $M_{\mathrm{ff}} \pm$ CI$_{95}$ [mN m] COSH	COSH$_{\mathrm{wide}}$	COSH$_{\mathrm{CAD}}$	NLS	CPD$_{\mathrm{N}}$	CPD$_{\mathrm{R}}$	CUSUM
1	x	−87.2	−49.6 ±28.3	−52.2 ±28.1	−49.1 ±30.8	−61.7 ±39.1	−43.9 ±30.9	−36.1 ±45.4	−34.5 ±37.2
1	y	+344.4	+336.3 ±12.6	+334.8 ±12.8	+334.5 ±23.5	+341.4 ±41.1	+337.0 ±18.2	+324.6 ±24.2	+288.6 ±116.8
2	x	−451.4	−478.6 ±11.5	−479.3 ±11.3	−479.4 ±20.3	−474.8 ±23.9	−476.8 ±22.0	−445.4 ±131.1	−473.7 ±16.6
2	y	+312.7	+334.0 ±13.0	+331.4 ±13.4	+327.4 ±25.4	+303.3 ±35.8	+313.0 ±16.6	+303.7 ±14.6	+269.8 ±126.9
3	x	−166.2	−271.7 ±14.3	−272.4 ±14.5	−274.0 ±15.2	−269.9 ±23.5	−256.0 ±37.8	−146.3 ±149.0	−266.8 ±29.2
3	y	−99.2	−22.8 ±12.5	−23.1 ±12.3	−23.0 ±18.8	−33.5 ±30.7	−22.4 ±25.8	−29.6 ±31.5	−35.3 ±107.2
4	x	+210.7	+165.5 ±20.6	+166.1 ±20.6	+165.6 ±20.9	+155.9 ±27.9	+176.0 ±22.6	+113.2 ±69.0	+158.3 ±28.7
4	y	−75.8	−4.3 ±19.3	−5.1 ±19.4	−10.3 ±25.0	+20.1 ±47.2	+21.0 ±43.9	+19.0 ±43.1	+6.9 ±43.1
5	x	+344.6	+340.7 ±22.9	+338.1 ±21.9	+340.7 ±25.4	+317.4 ±34.7	+325.1 ±20.7	+352.5 ±118.3	+323.8 ±23.5
5	y	+344.0	+333.8 ±21.5	+338.9 ±22.0	+325.5 ±29.4	+351.0 ±53.0	+342.3 ±30.8	+346.0 ±30.1	+342.3 ±31.6

Table 9: The quantity actually compensated, $M_{\mathrm{ff}} = \pm\frac{1}{2}(M_+ + M_-)$, for the same runs. Changing the COSH constants moves it by at most 5.1 mN m (PNLS-centred versus wide box) and 13.4 mN m (CAD versus wide box), against magnitudes up to 479 mN m — the compensation is insensitive to which of the three constant choices is adopted.

full data sit in the tables: Table 8 compares the deliverable — the CoM offset from the pivot-free average (7) — against the load-cell ground truth; Table 9 gives the feed-forward compensation moment itself; Table 10 lists the underlying directional critical-moment means, pivot-free offsets and their confidence intervals; and Table 11 compares the ramp-rate dispersion of the directional critical moments.

Three observations. *(i) The deliverable is detector-agnostic:* against the load-cell truth all five estimators land at RMS 1.6–1.8 mm per component, equivalently 2.3–2.6 mm radial (Table 8, bottom rows) — the pivot-free average (7) cancels each method's largely common-mode onset bias between the + and − directions, and 5–7 runs per direction average the rest. That the final CoM offset does not depend on the detection methodology is itself a validation certificate — and Fig. 4 sharpens it: the residual errors that do remain (cases 3–4) are shared by all five estimators, placing them in the vehicle–contact geometry rather than in any onset detector. *(ii) Run-level robustness is where the methods separate:* COSH has the lowest CV in 16/20 direction groups (CPD$_{\mathrm{N}}$ takes 3, CPD$_{\mathrm{R}}$ 1), median 2.5% against 4.1–7.1%, and its worst group is 5.7% where the alternatives reach 8.6–27.4% — the nonparametric detectors (CPD$_{\mathrm{R}}$, CUSUM) occasionally misplace the onset badly on short fast-ramp windows. Equivalently, the 95% confidence interval on the CoM offset is ±0.3–0.9 mm under COSH — the narrowest in 9 of 10 case–axis pairs — against up to ±4–5 mm for the nonparametric detectors (Fig. 3). The paired per-run critical moments (NLS vs. COSH) differ by 33 mN m at the median but up to 269 mN m — the ridge of Sec. 7 realized

method		1x	1y	2x	2y	3x	3y	4x	4y	5x	5y
COSH	$\bar{M}_-$	−1.043	−0.423	−1.824	−0.675	−1.538	−0.983	−1.138	−0.999	−1.016	−0.633
	$\bar{M}_+$	+0.944	+1.096	+0.867	+1.343	+0.995	+0.938	+1.469	+0.991	+1.698	+1.301
	M_{ff}	−0.050	+0.336	−0.479	+0.334	−0.272	−0.023	+0.165	−0.004	+0.341	+0.334
	CI_{95}	±0.028	±0.013	±0.011	±0.013	±0.014	±0.013	±0.021	±0.019	±0.023	±0.022
$\mathrm{COSH}_{\mathrm{wide}}$	$\bar{M}_-$	−1.054	−0.421	−1.829	−0.675	−1.542	−0.983	−1.136	−0.998	−1.019	−0.645
	$\bar{M}_+$	+0.950	+1.090	+0.870	+1.338	+0.997	+0.937	+1.468	+0.988	+1.695	+1.323
	M_{ff}	−0.052	+0.335	−0.479	+0.331	−0.272	−0.023	+0.166	−0.005	+0.338	+0.339
	CI_{95}	±0.028	±0.013	±0.011	±0.013	±0.014	±0.012	±0.021	±0.019	±0.022	±0.022
$\mathrm{COSH}_{\mathrm{CAD}}$	$\bar{M}_-$	−0.992	−0.372	−1.787	−0.597	−1.552	−0.959	−1.128	−0.949	−0.963	−0.584
	$\bar{M}_+$	+0.894	+1.041	+0.828	+1.252	+1.004	+0.913	+1.459	+0.928	+1.644	+1.235
	M_{ff}	−0.049	+0.334	−0.479	+0.327	−0.274	−0.023	+0.166	−0.010	+0.341	+0.326
	CI_{95}	±0.031	±0.024	±0.020	±0.025	±0.015	±0.019	±0.021	±0.025	±0.025	±0.029
NLS	$\bar{M}_-$	−1.018	−0.373	−1.792	−0.641	−1.573	−0.990	−1.151	−0.928	−0.989	−0.539
	$\bar{M}_+$	+0.894	+1.056	+0.843	+1.248	+1.033	+0.923	+1.463	+0.969	+1.624	+1.241
	M_{ff}	−0.062	+0.341	−0.475	+0.303	−0.270	−0.033	+0.156	+0.020	+0.317	+0.351
	CI_{95}	±0.039	±0.041	±0.024	±0.036	±0.024	±0.031	±0.028	±0.047	±0.035	±0.053
$\mathrm{CPD}_{\mathrm{N}}$	$\bar{M}_-$	−1.048	−0.456	−1.852	−0.707	−1.573	−1.059	−1.164	−1.005	−1.033	−0.647
	$\bar{M}_+$	+0.960	+1.130	+0.898	+1.333	+1.061	+1.014	+1.516	+1.047	+1.683	+1.332
	M_{ff}	−0.044	+0.337	−0.477	+0.313	−0.256	−0.022	+0.176	+0.021	+0.325	+0.342
	CI_{95}	±0.031	±0.018	±0.022	±0.017	±0.038	±0.026	±0.023	±0.044	±0.021	±0.031
$\mathrm{CPD}_{\mathrm{R}}$	$\bar{M}_-$	−1.023	−0.455	−1.680	−0.706	−1.339	−1.055	−1.179	−0.986	−0.932	−0.637
	$\bar{M}_+$	+0.950	+1.105	+0.789	+1.314	+1.046	+0.995	+1.406	+1.024	+1.637	+1.328
	M_{ff}	−0.036	+0.325	−0.445	+0.304	−0.146	−0.030	+0.113	+0.019	+0.353	+0.346
	CI_{95}	±0.045	±0.024	±0.131	±0.015	±0.149	±0.032	±0.069	±0.043	±0.118	±0.030
CUSUM	$\bar{M}_-$	−1.046	−0.476	−1.861	−0.714	−1.608	−1.029	−1.181	−1.044	−1.048	−0.667
	$\bar{M}_+$	+0.977	+1.054	+0.913	+1.254	+1.075	+0.958	+1.498	+1.058	+1.695	+1.351
	M_{ff}	−0.035	+0.289	−0.474	+0.270	−0.267	−0.035	+0.158	+0.007	+0.324	+0.342
	CI_{95}	±0.037	±0.117	±0.017	±0.127	±0.029	±0.107	±0.029	±0.043	±0.024	±0.032

Table 10: Directional critical-moment means $\bar{M}_\pm$, pivot-free offset M_{ff} and its 95% confidence interval (Welch t) [N m] per estimator. Columns are case–axis pairs (e.g. 1x = case 1, roll). The three COSH variants track each other closely, while NLS magnitudes sit low (early onsets) and $\mathrm{CPD}_{\mathrm{N}}$/CUSUM high (detection lag); M_{ff} stays nearly method-invariant throughout.

run by run. *(iii) The biases carry signatures:* NLS sits 3.7% *low* in mean $|M|$ (the early-onset drift of the free amplitude–onset fit), while $\mathrm{CPD}_{\mathrm{N}}$ and CUSUM sit 3.0% and 3.1% *high* — detection lag, since a distribution change must accumulate post-onset evidence before it is declared; both signatures largely cancel in (7). COSH needs no per-run tuning (CUSUM's allowance k, CPD's minimum segment) and a *single* run already sits within a few percent of its directional mean — which is what enables run-count reduction and per-run gating diagnostics.

10 The ground-induced moment at the tip-over threshold

The rigid tip-over balance ties each directional critical moment to quantities measured independently of the identification, $M_\pm = \pm(W - f)\, l_{\mathrm{arm}} + W y_{\mathrm{off}}$ (roll) and $M_\pm = \pm(W - f)\, l_{\mathrm{arm}} - W x_{\mathrm{off}}$ (pitch): W and the offsets from the load-cell truth, f the collective thrust reconstructed from the rotor speeds at the onset, and the pivot arms fitted from the *position* trace, a circle fit of the odometry motion about the contact line per run (`estimate_pivot_from_mocap`). Each run is predicted with its own arm and onset thrust, and the predictions are averaged per case/axis/direction. The kinematic arms come out consistent — 140.4 ± 2.3 mm (roll) and 112.7 ± 6.6 mm (pitch) across the five configurations, against the geometric 140/110 mm — and no simulation enters anywhere.

That balance does not close, and the gap is the subject of this section. Writing

$$\Delta M \triangleq T_{\mathrm{rigid}} - M_{\mathrm{det}}, \qquad T_{\mathrm{rigid}} = \mathrm{sgn}\,(W - f)\, l + S_{\mathrm{off}}, \tag{8}$$

the vehicle tips at a *smaller* commanded moment than the rigid balance requires, in all twenty direction groups, by 204 mN m in the median — about a fifth of the threshold itself. Something the rigid balance omits is helping it over.

One qualification belongs here rather than at the end. ΔM is not a measurement: it is what the rigid balance leaves over, and two of that balance's ingredients are themselves modelled. The collective thrust

		COSH		$\mathrm{COSH_{wide}}$		$\mathrm{COSH_{CAD}}$		NLS		$\mathrm{CPD_N}$		$\mathrm{CPD_R}$		CUSUM	
case	axis	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+
1	x	3.9	5.8	**3.8**	5.7	5.6	**5.5**	6.4	7.7	4.4	6.1	7.3	8.6	5.9	6.7
1	y	4.4	**2.2**	**4.2**	2.3	12.3	3.2	17.1	7.2	5.0	3.3	7.0	4.4	4.8	24.0
2	x	1.3	**1.6**	**1.2**	1.6	2.3	2.9	2.5	4.2	2.5	2.2	15.5	22.1	1.8	2.4
2	y	1.6	2.1	**1.5**	2.1	8.1	3.0	7.1	5.8	3.1	2.5	3.3	**2.0**	3.0	21.9
3	x	**1.7**	**2.3**	1.7	2.4	1.7	2.5	2.4	4.1	4.9	4.2	23.9	8.5	3.1	4.7
3	y	**1.7**	2.6	1.7	**2.6**	2.9	3.9	3.2	7.0	4.4	4.1	4.4	6.0	12.4	22.8
4	x	3.3	2.2	3.4	2.2	**3.1**	2.5	3.8	3.5	4.0	**1.6**	5.6	10.3	3.9	3.5
4	y	**2.8**	**3.7**	2.9	3.7	4.3	4.8	10.3	6.1	7.5	7.2	7.8	6.9	6.7	7.3
5	x	3.8	2.4	3.9	2.1	5.6	**1.5**	7.2	2.6	**3.6**	2.0	27.4	3.0	4.0	2.3
5	y	5.3	**3.1**	**5.3**	3.1	7.1	4.6	19.8	5.4	8.6	3.8	8.4	3.8	8.3	4.0
median / worst [%]		2.5 / 5.8		2.5 / 5.7		3.5 / 12.3		5.9 / 19.8		4.1 / 8.6		7.1 / 27.4		4.7 / 24.0	

Table 11: Ramp-rate dispersion (CV [%]) of the directional critical moments. Bold marks the lowest CV in each of the 20 direction groups: COSH in 8, $\mathrm{COSH_{wide}}$ in 6, $\mathrm{COSH_{CAD}}$ in 3, $\mathrm{CPD_N}$ in 2, $\mathrm{CPD_R}$ in 1. The underlying directional means are in Table 10.

	|residual| [mN m]				magnitude deficit	
thrust model	median	p90	max	RMS	mean [mN m]	one-sided
(a) no ground effect	204.8	301.5	349.5	213.3	−199.4	20/20
(b) GE, no rotor interference	163.9	258.9	304.2	176.1	−159.1	20/20
(c) GE with rotor interference	**61.0**	**127.6**	**177.4**	**86.0**	**−35.8**	12/20

Table 12: Residuals of the identified directional means against the static thresholds (odometry-fitted arms held fixed). The magnitude deficit is $|M_{\mathrm{ident}}| - |M_{\mathrm{pred}}|$, whose sign exposes systematic over-prediction (the raw signed residual hides it, the two tip directions carrying opposite signs); one-sided” counts the groups over-predicted. Each added physics element removes part of a strictly one-sided bias, and the parameter-free interference model leaves no one-sidedness worth the name.

is $\sum C_T \omega_i^2$ with C_T from a load-cell bench, so it is an out-of-ground-effect value by construction; and the circle fit locates the *kinematic* pivot, whereas the balance needs the line of action of the normal-force resultant, which need not lie there. ΔM therefore collects everything the ground does that those two ingredients miss — aerodynamic and contact alike — and the question is how much of it near-ground aerodynamics accounts for.

How much the model accounts for. Ground effect enters through *both* channels of the pivot-moment decomposition $\Delta M_{\mathrm{GE}} = \mathrm{sgn} \cdot a + b\,M$ (`analysis/ge_linearity.py`): the thrust channel $a = c_a f l$ and the moment-proportional channel b. The onset balance $M(1+b) + \mathrm{sgn}\, a = T_{\mathrm{rigid}}$ gives

$$M_{\mathrm{pred}} = \frac{T_{\mathrm{rigid}} - \mathrm{sgn} \cdot c_a f l}{1+b}, \tag{9}$$

evaluated at three levels of physics: *(a)* no ground effect; *(b)* ground effect *without* rotor–rotor interference — the per-rotor Cheeseman–Bennett superposition at the tilted heights ($c_a = 1.03\%$, $b = 1.03\%$); and *(c)* ground effect *with* rotor interference — the image superposition over the rotor-centre distances, $s_{\mathrm{rot},j} = (R^2/4) \sum_n (z_j + z_n)/[d_{jn}^2 + (z_j + z_n)^2]^{3/2}$, $\gamma_j = (1 - s_{\mathrm{rot},j})^{-1}$, whose self term reduces exactly to Cheeseman–Bennett and which contains *no empirical constant of any kind* ($c_a = b = 4.31\%$).

Evaluated at the onset, that model predicts 165 mN m of ground-induced moment per direction against the 204 the gap implies: it accounts for 81% of it, with nothing tuned. Neglecting rotor–rotor interference accounts for only a fifth. Group by group the ratio of implied to modelled ranges from 0.53 to 2.07, so the agreement is a statement about the median and not about any single group.

A consistency check that needs no truth. Eq. (35) estimates the same offset twice, once from each tip direction, using that direction's own arm, and Eq. (34) supplies the weight from the moments themselves. Both are internal: no load cell and no external reference enters. Neither closes without a ground-effect term. The two directional estimates of the offset disagree by 12.6 mm in the median and 19.8 at worst — that is the antisymmetric part of (8) expressed in millimetres — and Eq. (34) returns a weight 5.2% below mg, low in all ten configurations. Restoring the interference channels

			l_{odom}	COSH	no GE		no int.	with interference		benchmark
case	axis	dir	[mm]	$\bar{M}$ [N m]	pred	resid	resid	pred	resid	range [N m]
1	x	-	145.2	−1.051	−1.401	+350	+304	−1.217	+165	[−1.048, −1.018]
1	x	+	139.2	+0.950	+1.172	−222	−180	+1.002	−52	[+0.894, +0.977]
1	y	-	109.3	−0.421	−0.630	+209	+179	−0.508	+87	[−0.476, −0.373]
1	y	+	120.1	+1.090	+1.431	−341	−301	+1.268	−177	[+1.054, +1.130]
2	x	-	140.0	−1.829	−1.933	+104	+54	−1.731	−98	[−1.861, −1.680]
2	x	+	139.4	+0.870	+1.019	−149	−109	+0.856	+14	[+0.789, +0.913]
2	y	-	105.0	−0.675	−0.791	+116	+86	−0.667	−8	[−0.714, −0.641]
2	y	+	118.5	+1.338	+1.566	−228	−187	+1.398	−60	[+1.248, +1.333]
3	x	-	142.5	−1.542	−1.672	+130	+82	−1.479	−63	[−1.608, −1.339]
3	x	+	138.3	+0.997	+1.294	−297	−254	+1.121	−123	[+1.033, +1.075]
3	y	-	109.3	−0.984	−1.253	+270	+233	−1.106	+123	[−1.059, −0.990]
3	y	+	117.9	+0.937	+1.145	−208	−171	+0.995	−58	[+0.923, +1.014]
4	x	-	141.1	−1.136	−1.280	+144	+101	−1.104	−32	[−1.181, −1.151]
4	x	+	138.4	+1.468	+1.674	−206	−159	+1.484	−16	[+1.406, +1.516]
4	y	-	106.7	−0.998	−1.202	+204	+169	−1.060	+62	[−1.044, −0.928]
4	y	+	118.3	+0.982	+1.174	−192	−154	+1.022	−40	[+0.969, +1.058]
5	x	-	142.5	−1.019	−1.158	+140	+97	−0.987	−32	[−1.048, −0.932]
5	x	+	137.9	+1.695	+1.804	−109	−61	+1.609	+86	[+1.624, +1.695]
5	y	-	103.2	−0.645	−0.741	+96	+66	−0.620	−25	[−0.667, −0.539]
5	y	+	118.9	+1.323	+1.601	−278	−236	+1.432	−109	[+1.241, +1.351]

Table 13: Per-direction static-threshold check with odometry-fitted arms; residuals in mN m. The last column pair is the parameter-free rotor-interference model; the rightmost entry of each row is the range spanned by the four alternative estimators (NLS, $\mathrm{CPD_N}$, $\mathrm{CPD_R}$, CUSUM), showing that the disagreement with the static prediction is not an artifact of the onset detector.

brings the directional spread to 3.6 mm (worst 9.0), a factor of 3.5, and the weight to 1.1% below (`analysis/offset_pivot_based.py`). The thrust channel does most of the second, since it enters Eq. (34) as $(1 + c_a)f$.

Neither result identifies ground effect. A 19 mm inboard shift of the contact resultant closes both, exactly as it closes the threshold deficit, and the degeneracy of the next paragraphs applies here unchanged. What they add is that one parameter-free correction of the right size removes an inconsistency that shows up in three independent places — the directional thresholds, the directional offsets, and the weight — none of which was used to choose it.

It is not the onset detector. The four alternative estimators bracket the identified moment within a median 108 mN m (last column of Table 13), and all seven detectors of Sec. 9 place the per-run onset moment within 3–45 mN m of one another. Both are well below the gap, so ΔM is a property of the contact physics and not of where the change point is put. The interference prediction falls inside the estimator bracket in 9 of the 20 groups, where neither of the other two levels ever does.

What the remainder is. Normalised by each group's own threshold, the three levels leave 18.4%, 14.5% and 5.2% in the median and 49.0%, 41.9% and 20.2% at their worst. So the reported model still misses each directional threshold by about a twentieth of its value. Of that residual 97% is systematic within a direction group (57.0 against 12.4 mN m of run noise carried into the group mean), so it is not detection scatter and no number of repeats removes it.

A contact-geometry explanation covers it, and covers the whole gap equally well if allowed to: if the normal-force resultant acts a distance δl *inboard* of the kinematic pivot circle — as a finite, compliant contact patch implies — the least-squares fit gives $\delta l = 19.2 \pm 1.8$ mm with a residual RMS of 82.3 mN m, statistically indistinguishable from the 86.0 mN m of the ground-effect hypothesis (`analysis/static_attribution.py`). The two are degenerate in a single group because both act as force × length.

The five configurations break part of that degeneracy. The airframe, the rotor heights and the collective thrust are identical in all of them and only ballast moves horizontally, so any aerodynamic term is common across cases and can contribute nothing to the case-to-case spread. Splitting the antisymmetric residual accordingly, the varying part is 73.3 mN m sd on roll and 40.7 on pitch — 6.9 and 3.9 mm of contact lever between configurations, with no aerodynamic explanation available. The common part fails a sharper test: a body-level vertical force, whether the fountain lift the model excludes, an error in the collective thrust or one in C_T, reaches the threshold through the same lever as the weight and must therefore show 1.27× on roll what it shows on pitch. Pitch carries -69.8 ± 18.2 mN m and roll -6.7 ± 32.8, where that hypothesis demands -88.9: wrong by 2.5 standard errors and in the wrong direction, ten times larger on the axis where it should be smaller (`analysis/static_residual_split.py`). Roll therefore carries no significant axis-common term once the interference model is applied; pitch carries one, and it is specific to that axis.

The floor. The contact arm sets how far this can be pushed. The threshold's sensitivity to it is $\mathrm{d}M/\mathrm{d}l = W - f = 10.6$ N, so the 61 mN m median residual is 5.8 mm of arm, the same order as the run-to-run spread of the circle fit inside a single group (0.7–8.4 mm, median 1.5) and smaller than any plausible systematic on where the resultant acts within a compliant patch. Nothing finer about the aerodynamics is resolvable here, which is why this section reports a budget and not a fit. Breaking the degeneracy needs an experiment this one does not contain: the ground-effect term scales with h/R and the contact term does not, so repeating a single axis at a second clearance separates them — at $h = 0.60$ m the aerodynamic hypothesis predicts 0.069 N m against the contact hypothesis's unchanged 0.20, a three-fold split well above the 0.06 N m arm floor.

Why none of this reaches the deliverable. Write the unmodelled part of the balance as a lumped $\Delta_\pm$, whatever its mechanism, and split it by direction, $\Delta_a = (\Delta_+ - \Delta_-)/2$ and $\Delta_s = (\Delta_+ + \Delta_-)/2$. Everything that acts as force × pivot-distance changes sign with the tip direction and lands in Δ_a, and

$$M_{\mathrm{ff}} = \tfrac{1}{2}\,(M_+ + M_-) = \tfrac{1}{2}(W - f)(l_+ - l_-) + S_{\mathrm{off}} + \Delta_s, \tag{10}$$

so Δ_a cancels identically — the ground-effect thrust channel, a common displacement of the contact resultant and rolling resistance with it. Measured here, $|\Delta_a| = 54$ mN m in the median and $|\Delta_s| = 37$, the latter 3.7% of the half-width of the interval $[M_-, M_+]$ the excitation measures directly (1000 mN m in the median, 8.2× the worst $|\Delta_s|$). Since any moment strictly inside that interval keeps both contacts loaded, and the midpoint maximises the smaller of the two margins, M_{ff} is the max–min-margin choice whatever $\Delta_\pm$ turns out to be; both endpoints are obtained from the excitation, not from a model.

The moment applied, and the offset reported. The two uses of the same measurement need separating, because the ground-effect term belongs in one and not the other. The compensation applied at take-off is M_{ff} itself, with no ground-effect model anywhere: it is the in-situ balanced moment and already carries whatever the ground contributes, which is precisely what has to be cancelled. Subtracting a modelled ground-effect term there would remove part of the disturbance the compensation exists to reject.

Reading the same quantity as a mass property is a different matter, and it is the one the reviewer's concern touches: the load cell measures $W\lambda_{\mathrm{off}}$ alone, so a ground-induced moment during the excitation would bias the inferred offset. The inversion that answers this restores both channels in the onset balance and pairs the directions,

$$S_{\mathrm{off}} = \tfrac{1+b}{2}\,(M_+ + M_-) + \tfrac{c_a}{2}\,(f_+ l_+ - f_- l_-) - \tfrac{1}{2}\big[(W - f_+)l_+ - (W - f_-)l_-\big], \tag{11}$$

the last bracket being dropped in the pivot-free variant, which assumes $l_+ = l_-$. Nothing in (11) is fitted: c_a and b are the model values at $\varphi = 0$. Table 14 evaluates it both ways.

The ground-effect term does not improve the offset. Held at the pivot-free estimator the paper reports, restoring it moves the RMS from 1.64 to 1.82 mm and the mean error from -0.91 to -1.20; with the arms restored as well it moves the RMS from 1.77 to 1.65 and the mean from $+0.42$ to $+0.13$. No paired comparison reaches significance — the correction is 0.41 mm against a 0.45 mm standard error — so the honest reading is that this dataset cannot resolve the ground-effect contribution to the offset at all, and that its size is bounded at about half a millimetre either way. What does change systematically is the mean bias, which the fullest inversion removes ($-0.91 \to +0.13$ mm) at unchanged RMS, leaving the scatter that the per-configuration contact geometry accounts for.

The offset is therefore reported at $\eta^{GE} = 0$, matching the quantity applied in flight, with Table 14 standing as the bound on what the omission costs. This is why M_{ff} validates to millimetres (Sec. 9) even though the individual thresholds carry a $\sim$0.2 N m ground-induced budget whose internal split this experiment cannot resolve.

Table 14: Identified CoM offset against the load-cell truth, with and without the ground-effect channels of (11), over the ten case/axis configurations. "Δ vs $\eta = 0$" is the mean paired change in |error|, with the exact sign test beside it. The standard error of the comparison is 0.45 mm and the interference correction is 0.41 mm, so none of the differences is resolvable (`analysis/offset_ge_variants.py`).

arms	ground effect	RMS [mm]	mean [mm]	max [mm]	Δ vs η=0 [mm]	p
pivot-free	none (reported)	1.64	−0.91	3.34	—	—
pivot-free	single-rotor	1.67	−0.98	3.44	−0.00	1.00
pivot-free	interference	1.82	−1.20	3.77	+0.19	0.11
pivot-based	none	1.77	+0.42	2.96	—	—
pivot-based	single-rotor	1.73	+0.35	2.80	−0.01	1.00
pivot-based	interference	1.65	+0.13	3.05	−0.06	1.00

11 Reproducibility

Every number in this document is reproducible from the repository:

- `critical_value_getter_piecewise.py` — pipeline, gates, caps, calibration, Algorithm 1;
- `analysis/ramp_quality.py` — gates;
- `analysis/sensitivity.py` — ±20% robustness;
- `analysis/cosh_fidelity.py` — prediction-test residuals;
- `analysis/rate_residual.py` — Table 3, endpoint and window residuals against the deviation bound;
- `analysis/excitation_angle_design.py` — rate/cutoff design;
- `analysis/ge_linearity.py`, `analysis/ge_trajectory.py` — ground-effect bracket;
- `analysis/tiltcap_ablation.py` — window ablation;
- `analysis/baseline_stat_ablation.py` — median/mean/zero baseline;
- `analysis/bias_drift.py` — in-window baseline drift;
- `analysis/nls_comparison.py` — Tables 8–11;
- `analysis/mcrit_prediction.py`, `analysis/mcrit_static_figure.py`, `analysis/offset_pivot_based.py`, `analysis/offset_ge_variants.py`, `analysis/static_residual_split.py`, `analysis/static_attribution.py` — Sec. 10 (static-threshold budget, arms, attribution limits, Fig. 5);
- `analysis/ge_bilinear.py` — the bilinear $\tau\varphi$ ground-effect term along measured trajectories;
- `analysis/constrained_calibration.py` — Remark 2 and Table 6 (the shared $(Wz_{\mathrm{CoM}}, J_{P,\mathrm{roll}}, J_{P,\mathrm{pitch}})$ reparametrization and the rerun identification);
- `analysis/cv_table.py` — Table 4 (rate consistency under the free, CAD and mis-specified constants);
- `analysis/phys_compare.py` — Table 5 (calibrated constants against the CAD reference and the parallel-axis floor);
- `analysis/cad_reference.py` — Table 7 (the identification rerun with nothing fitted);
- `analysis/pivot_geom.py` — the mocap arc fit for l_p and the marker height;
- `analysis/pnls_centred.py`, `analysis/pnls_ci.py` — the two-stage calibration (per-run PNLS medians, then the score refined in a neighbourhood of them) and its intervals;
- `analysis/mff_table.py`, `analysis/cv_full.py` — the estimator benchmark extended with the PNLS-centred and CAD constants, and the full-pipeline dispersions;

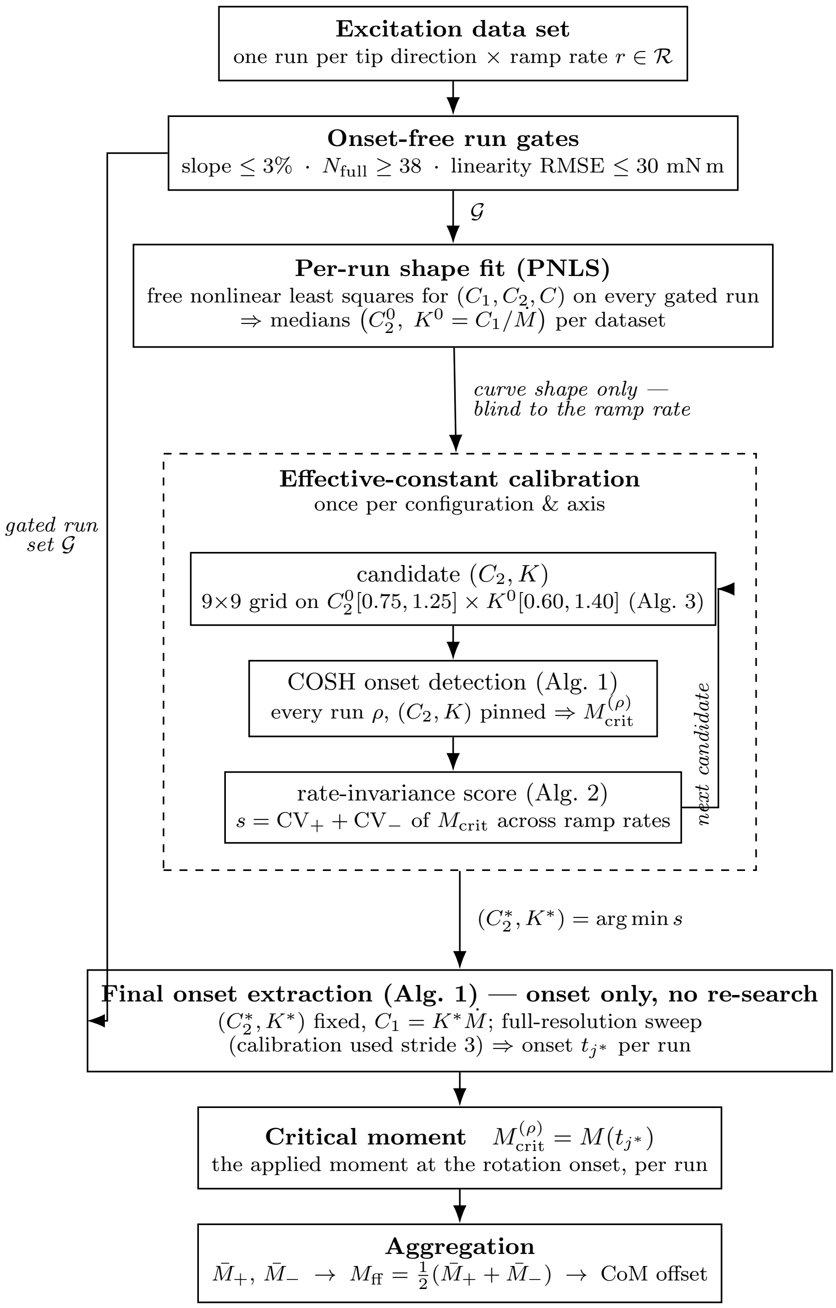

Figure 1: Pipeline at a glance. The dashed container is the once-per-configuration calibration: it iterates candidate pairs (C_2, K), running the *same* onset detector (Algorithm 1) on every gated run and scoring the candidates by rate-invariance of the resulting critical moments. The selected pair then drives one final onset extraction per run — the constants are *not* re-estimated; only the onset index is re-swept, at full resolution, because the calibration subsamples the sweep by a stride of 3 for speed — whose onset sample yields $M_{\text{crit}} = M(t_{j^*})$, and the directional means aggregate into the pivot-free feed-forward moment.

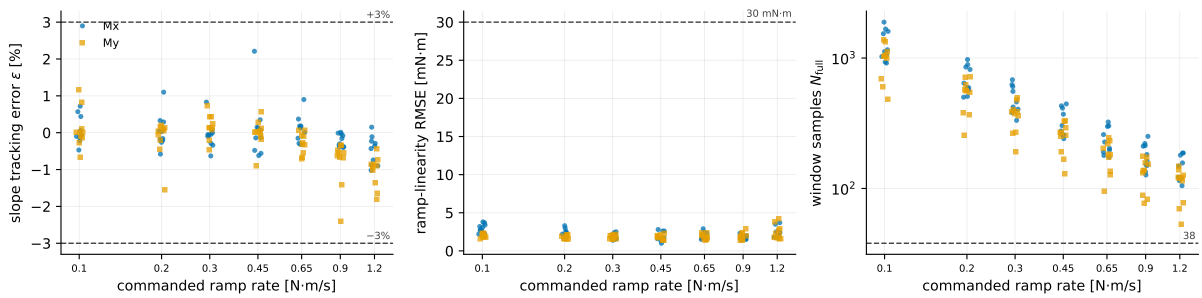

Figure 2: Run-gate quantities for all 140 runs versus the commanded ramp rate, in gate order: slope tracking error ε and ramp-linearity RMSE on the floored segment ($|M| \geq M_{\mathrm{floor}}$), and the excitation-window sample count N_{full}. Dashed lines are the gate thresholds (±3%, 30 mN m, 38 samples); every run clears all three.

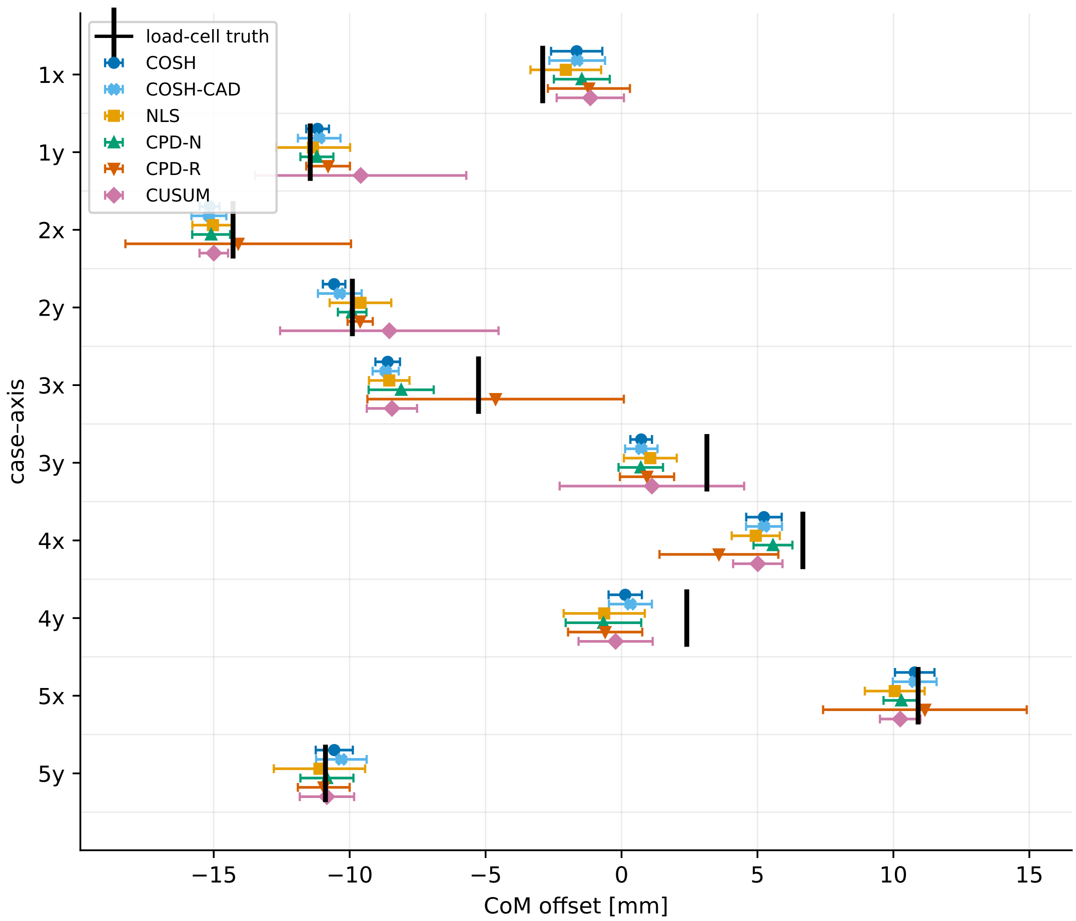

Figure 3: Estimated CoM offset with 95% confidence intervals for six estimators, against the load-cell truth (black tick), per case–axis pair. COSH (dark blue) has the narrowest interval in 9 of 10 pairs (±0.36–0.94 mm); COSH$_{\mathrm{CAD}}$ (light blue) tracks it closely in the offset while carrying a wider interval, which is the rate-consistency penalty of Table 11. The nonparametric detectors blow up to ±4–5 mm on the pairs where they misplace onsets (CPD$_{\mathrm{R}}$ at 2x, 3x, 5x; CUSUM at 1y, 2y, 3y). COSH$_{\mathrm{wide}}$ is omitted here because it is visually indistinguishable from COSH; it appears in the tables. Marker shapes duplicate the color coding.

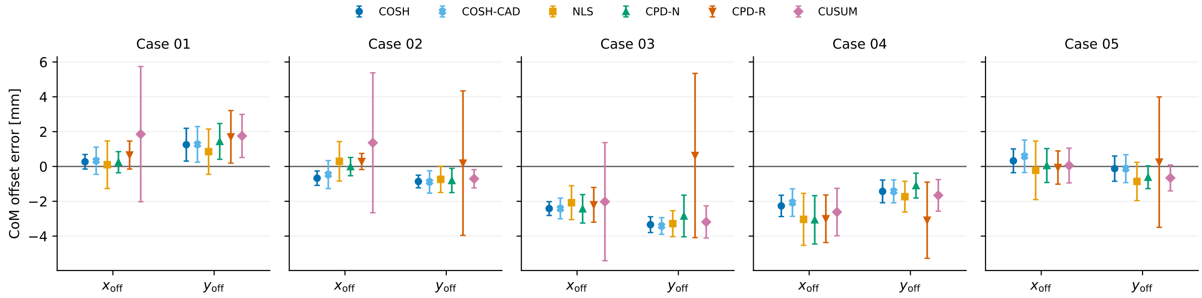

Figure 4: The same data folded about the truth: CoM offset *error* (estimate − truth) with 95% confidence intervals, grouped by component within each case (x_{off} from the pitch excitation, y_{off} from the roll excitation). The residual biases of cases 3 and 4 are *common to every estimator* — a vehicle–contact asymmetry, not a detection artifact — while the estimator-to-estimator differences stay within about ± 1 mm.

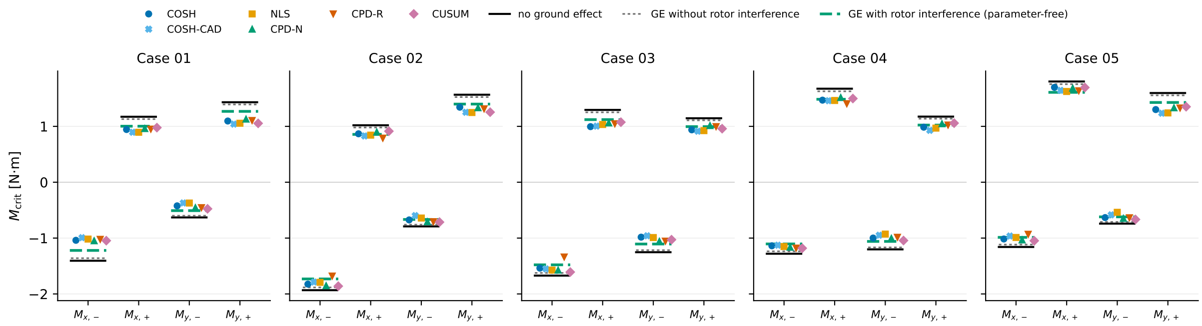

Figure 5: Identified directional critical moments (five estimators, colored markers) against the static-threshold predictions with truth offsets and odometry-fitted arms: no ground effect (black solid), ground effect without rotor interference (gray dotted), and with rotor interference (green dashed). The interference prediction falls inside the range spanned by the five estimators in 9 of the 20 groups, where neither of the other two ever does (0/20); the median estimator range is 108 mN m. A 19 mm inboard shift of the static contact lever would move the black line by a similar amount: the figure shows the *size* of the near-ground correction, not its attribution.